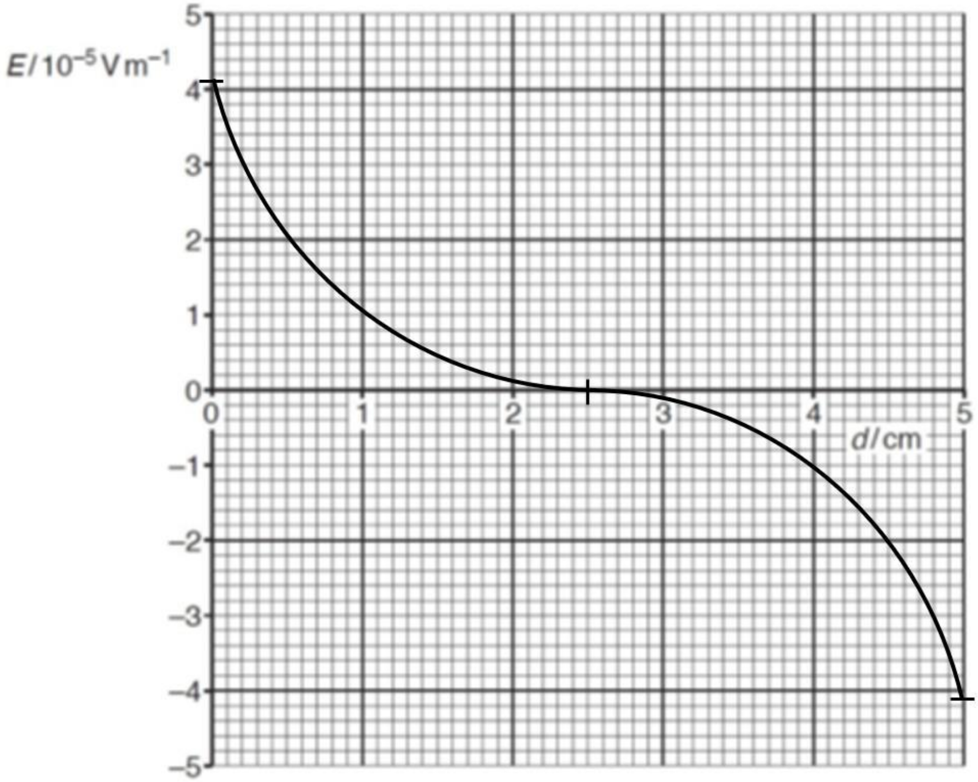
6 **(a)** $E = Q / 4\pi\varepsilon_0 r^2$ or $E = kQ / r^2$ with k defined / substituted in

$4.1 \times 10^{-5} = [Q / (4\pi \times 8.85 \times 10^{-12} \times 0.025^2)] - [Q / (4\pi \times 8.85 \times 10^{-12} \times 0.075^2)]$ M1

$Q = 3.2 \times 10^{-18}$ C A1

(b)



correct shape (start positive E, gradient trend) B1

through points (0, 4.1×10^{-5}) (2.5, 0) & (5.0, - 4.1×10^{-5}) B1

(c) Using the graph, E-field strength on the left of d = 2.5 cm is positive, while on the right is negative. Thus, the graph shows E-field is always directed towards the d = 2.5 cm. B1

Electric Force, F = qE, (or acceleration) on positive charge is always opposite to displacement from d = 2.5 cm B1

7 **(c)** V = 3.4 V = V